Chapter 6. Alternative optimization methods: Evolutionary algorithms

This chapter covers

- Evolution algorithms for solving optimization problems
- Pros and cons of evolutionary approaches versus previous algorithms
- Solving the CartPole game without backpropagation
- Why evolutionary strategies can scale better than other algorithms

Neural networks were loosely inspired by real biological brains, and convolutional neural networks were also inspired by the biological mechanism of vision. There is a long tradition of advances in technology and engineering being motivated by biological organisms. Nature, through the process of evolution by natural selection, has solved many problems elegantly and efficiently. Naturally, people wondered whether evolution itself could be borrowed and implemented on a computer to generate solutions to problems. As you will see, we can indeed harness evolution to solve problems, and it works surprisingly well and is relatively easy to implement.

In natural evolution, biological traits change and new traits are generated simply by the fact that some traits confer a survival and reproduction advantage that results in those organisms being able to seed more copies of their genes in the next generation. The survival advantage of a gene depends entirely on the environment, which is often unpredictable and dynamic. Our use cases for simulated evolution are much simpler, since we generally want to maximize or minimize a single number, such as the loss when training a neural network.

In this chapter you will learn how to use simulated evolutionary algorithms to train neural networks for use in reinforcement learning without using backpropagation and gradient descent.

6.1. A different approach to reinforcement learning